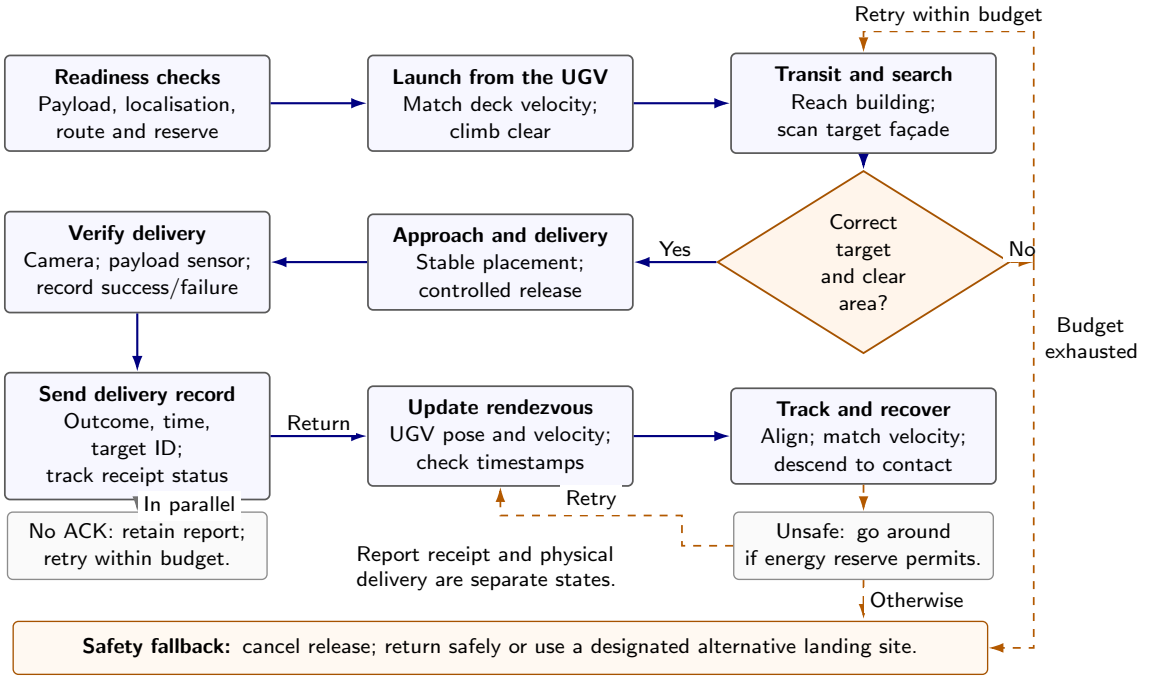
Retry within budget
Readiness checks
Payload, localisation, route and reserve
Launch from the UGV
Match deck velocity; climb clear
Transit and search
Reach building; scan target façade
Correct target and clear area?
Yes
No
Approach and delivery
Stable placement; controlled release
Verify delivery
Camera; payload sensor; record success/failure
Budget exhausted
Send delivery record
Outcome, time, target ID; track receipt status
Return
Update rendezvous
UGV pose and velocity; check timestamps
Track and recover
Align; match velocity; descend to contact
In parallel
No ACK: retain report; retry within budget.
Retry
Unsafe: go around if energy reserve permits.
Report receipt and physical delivery are separate states.
Otherwise
Safety fallback: cancel release; return safely or use a designated alternative landing site.